

# Juan David Munoz Bolanos

Junior Optical System Engineer

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+43 676 9115145

jmunozbolanos@gmail.com



## Summary

Physicist and Optical Systems Engineer with extensive experience in the end to Polishing complex optomechatronic and medical devices from concept to series. Proven track record in translating physical concepts into robust prototypes using SolidWorks, structured V model systems engineering, and rigorous testing protocols. Adept at collaborating in interdisciplinary teams to deliver high quality technical documentation and system integration for clinical and industrial applications.

## Research/Professional Experience

Jan 2022 to Jun 2026      **Medical University of Innsbruck**      *Innsbruck, Austria*  
**Optical Systems Engineer / PhD Researcher**

**End to End System Design:** Led the complete hardware to software build of a multiphoton microscope, integrating laser sources, galvo scanners, and adaptive optics. Achieved diffraction limited resolution.

**Systems Engineering:** Structured instrument development following V model principles, managing requirements tracking from concept to component testing. Utilized FMEA and 8D problem solving to prevent and resolve technical defects.

**Quality & Testing:** Executed structured Design of Experiments (DoE) and optical simulations in Python. Applied testing and documentation protocols for optomechanical assemblies with awareness of ISO 9001 and ISO 10110 frameworks.

**CAD & Prototyping:** Designed custom optomechanical mounts and alignment tools using SolidWorks and FreeCAD, ensuring precise mechanical tolerances and seamless integration.

Mar 2020 to Dec 2021      **Leibniz Institute of Photonic Technology**      *Jena, Germany*  
**R&D Research Assistant**

**Spectrometer Prototyping:** Designed and built a dispersive 1064 nm fiber probe Raman spectrometer for non invasive human tissue characterization, handling optical layout and detector calibration.

**Data Analysis & ML:** Developed a custom Python package (SpectraMap) to apply supervised machine learning algorithms to 2D hyperspectral Raman images, achieving high accuracy classification of tissue samples.

**Technical Documentation:** Authored comprehensive technical reports, system manuals, and peer reviewed publications detailing optical design parameters and calibration procedures.

## Technical Competencies

|                   |                                   |                                 |                    |
|-------------------|-----------------------------------|---------------------------------|--------------------|
| Optics & Hardware | Adaptive optics                   | Multiphoton/confocal microscopy | Raman spectroscopy |
|                   | Laser systems (fs, CW, 1064 nm)   |                                 | Camera integration |
|                   | 3D CAD (SolidWorks, FreeCAD)      |                                 |                    |
| Software & Tools  | Python (JAX, scikit-learn, NumPy) | LabVIEW/FPGA                    | C++                |
|                   | MATLAB                            |                                 | OpenCV             |
| Soft Skills       | Interdisciplinary teamwork        | Fast learner on new toolchains  |                    |
|                   |                                   |                                 |                    |

Education

|                      |                                                                                     |                    |
|----------------------|-------------------------------------------------------------------------------------|--------------------|
| Jan 2022 to Jun 2026 | Medical University of Innsbruck<br><i>Doctor of Philosophy in Optical Metrology</i> | Innsbruck, Austria |
| Oct 2019 to Sep 2021 | Friedrich Schiller Univ. Jena<br><i>Master of Science in Photonics</i>              | Jena, Germany      |
| Aug 2012 to Aug 2018 | University of Cauca<br><i>Bachelor of Science in Physics Engineering</i>            | Popayán, Colombia  |

Publications

- Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).
- Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).
- Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills

- Systems & Quality Engineering:** V-Model, 8D Problem Solving, Design of Experiments, ISO 9001.
- Adaptability & Independent Working:** Self-driven approach to bringing concepts to working systems; comfortable in dynamic, fast-paced environments.
- Teamwork & Communication:** Cross-functional collaboration across optics, hardware, software, and production teams; structured root cause analysis.

Personal Interests

- Biking/Hiking
- Bachata Dancing
- Harmonica
- Learning German

Languages

- English (Fluent)
- German (Intermediate B2)
- Spanish (Native)

References

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|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Prof. Dr. Monika Ritsch-Marte</b><br><i>Medical University of Innsbruck</i><br>monika.ritsch-marte@i-med.ac.at | <b>PD Dr. Christoph Krafft</b><br><i>Leibniz IPHT</i><br>christoph.krafft@leibniz-ipht.de |
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# **INTECOL**

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

## **TO WHOM IT MAY CONCERN**

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

## **INTECOL**

NIT. 811.041.410-4 - Tel.: 3163822

## **LENY ADRIANA RUIZ PINO**

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.  
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DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCES, OFFICIAL TRANSLATOR  
BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC  
OF COLOMBIA.

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**REPUBLIC OF COLOMBIA**  
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**CONSIDERING THAT**

**JUAN DAVID MUÑOZ BOLAÑOS**  
**CC 1.061.772.278 OF POPAYAN**

**HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,**  
**GRANTS THE TITLE OF**

**PHYSICAL ENGINEER**

**WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM**  
**FOR THE PROFESSIONAL EXERCISE**

**POPAYAN, 16 MARCH 2018**

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**RESOLUTION No. R-266-07 MINUTE No. 14-07**

**(SIGNED)**

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**THE DEAN OF THE FACULTY**  
**THE SECRETARY GENERAL OF THE UNIVERSITY**

*Juan Carlos Ojeda - Garces*  
TRADUCTOR E INTERPRETE OFICIAL  
RESOLUCIÓN No 1171  
MINISTERIO DE JUSTICIA 1997

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**20 DECEMBER 2018**





# URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die  
Physikalisch-Astronomische Fakultät  
mit dieser Urkunde

**Herrn Juan David Munoz Bolanos**  
**geb. 04.06.1994 in La Cruz (Kolumbien)**  
den Hochschulgrad

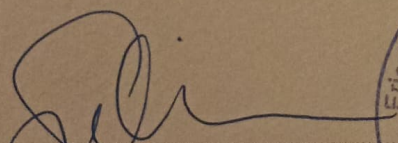
## MASTER OF SCIENCE (M.Sc.)

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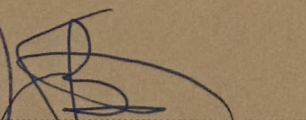
Er hat die Masterprüfung (120 ECTS)  
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021

  
Prof. Dr. Christian Spielmann  
Dekan



  
Prof. Dr. Silvana Botti  
Vorsitzende des  
Prüfungsausschusses

# Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

## ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

### bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r  
Maria VALLAZZA



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österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291



# Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

## ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

### bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r  
Maria VALLAZZA



ÖSD ist Mitglied von



# ösd

österreich schweiz deutschland



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